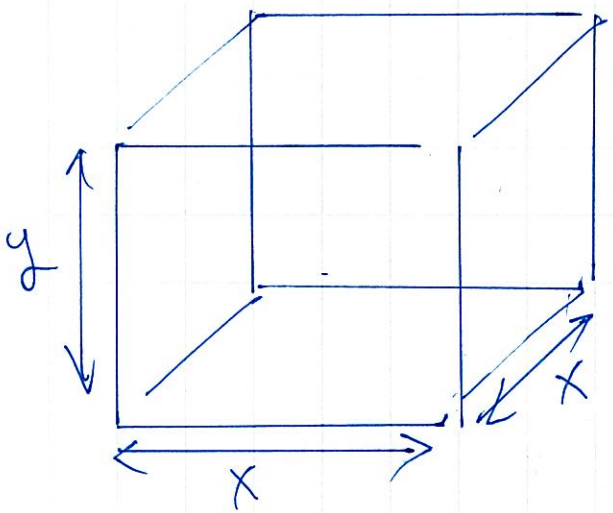
What do we mean by solving an optimization problem?

Let's design a box with largest possible volume such that
- the base of the box is a square
- the surface area of the box will be at most 500 cm²
- the box will not have a top.

Decision vars, objective func, constraints.



Decision vars: x = length/width of box
y = height of the box

Obj. function: $x^2 y$

Constraints: $x^2 + 4xy \leq 500$
$x \geq 0 \quad y \geq 0$

opt. problem $\max \; x^2 y$
st $x^2 + 4xy \leq 500$
$x \geq 0, \; y \geq 0$

According to Excel, the opt. solution is $(\hat{x}, \hat{y}) = (12.91, 6.45)$ with the opt volume 1075.82

- This optimization problem is nonlinear because either the obj. function is nonlinear in the decision vars, or the left sides of constraints are nonlinear in the decision vars or both.

- It turns out opt. software may very easily fail when we solve a nonlinear optimization problem

- E.g. Try to use Excel for the previous opt. problem with the initial guesses $x=0$, $y=0$.

Role of Optimization in Data Analysis, ML and AI

- Typical data set in an ML/AI application looks like $\{(x_j, y_j) : j = 1, \dots, m\}$, where $x_j \in \mathbb{R}^n$ is a vector of features and $y_j \in \mathbb{R}$ is the label corresponding to data point j.

- The goal is to find a prediction function $\phi(\cdot, w) : \mathbb{R}^n \to \mathbb{R}$ parameterized by $w \in \mathbb{R}^p$ such that $\phi(x_j, w) \approx y_j$ on a large fraction of data points

- Let $\ell(x_j, y_j, w)$ be the "loss" or "error" we incur when we use the prediction function parametrized by w to predict the the label for the j-th data point

- E.g for regression, $\ell(x_j, y_j, w) = (\phi(x_j, w) - y_j)^2$

- The opt. problem in ML/AI context looks like

$$\min_{w \in \mathbb{R}^p} \frac{1}{m} \sum_{j=1}^{m} \ell(x_j, y_j, w)$$

Least Squares Regression

- We predict the label for data point j as $w^t \cdot x_j$, where $w \in \mathbb{R}^n$ is the vector of parameters we need to estimate.

- In this case the opt. we solve is of the form

$$\min_{w \in \mathbb{R}^n} \frac{1}{m} \sum_{j=1}^{m} (w^t x_j - y_j)^2$$

- Ridge regression: $$\min_{w \in \mathbb{R}^n} \frac{1}{m} \sum_{j=1}^{m} (w^t x_j - y_j)^2 + \lambda \underbrace{\|w\|_2^2}_{\sum_{i=1}^{n} w_i^2}$$

useful for avoiding overfitting, ensuring robustness against outliers

- Lasso: $$\min_{w \in \mathbb{R}^n} \frac{1}{m} \sum_{j=1}^{m} (w^t x_j - y_j)^2 + \lambda \underbrace{\|w\|_1}_{\sum_{i=1}^{n} |w_i|}$$

useful for feature selection

Logistic Regression:

- Consider a two-class classification problem with the dataset $\{(x_j, y_j) : j = 1, \dots, m\}$, where $x_j \in \mathbb{R}^n$ and $y_j \in \{+1, -1\}$

- Our prediction function is of the form

$$\phi(x, w) = \frac{1}{1 + e^{w^t \cdot x}},$$

where $\phi(x, w)$ can be interpreted as the probability that a data point with feature x belongs to class $+1$

- $1 - \phi(x, w) = \dfrac{e^{w^t x}}{1 + e^{w^t x}}$ is the probability that a data point with feature vector x belongs to class -1

To fit a logistic regression model, we use maximum likelihood estimation

data point #	feature vector	label	likelihood
1	x_1	$+1$	$1/1+e^{w^t \cdot x_1}$
2	x_2	-1	$e^{w^t \cdot x_2}/1+e^{w^t \cdot x_2}$
3	x_3	$+1$	$1/1+e^{w^t \cdot x_3}$
$\vdots$	$\vdots$		$\vdots$
m	x_m	-1	$e^{w^t \cdot x_m}/1+e^{w^t \cdot x_m}$

Likelihood of the data set:

$$L(w) = \frac{1}{1+e^{w^t \cdot x_1}} \; \frac{e^{w^t \cdot x_2}}{1+e^{w^t \cdot x_2}} \; \frac{1}{1+e^{w^t x_3}} \cdots \frac{e^{w^t \cdot x_m}}{1+e^{w^t \cdot x_m}}$$